# Circuit Analysis — Lab 4

## Experiment 4: Implementation of Basic Parallel Circuit

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| **Name:** | ______________________________ | **Registration No:** | ______________________________ |
| **Date:** | ______________________________ | **Grade & Signature:** | ______________________________ |

**Objectives:** To make measurements in your own constructed parallel circuits.

**Equipment Required:** Power Supply, Breadboard, Connecting Wires, Resistances, Multimeter.

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## Lab Tasks

### Task 1: Parallel Circuit — Vs = 9V, R1 = 10 kΩ, R2 = 2 kΩ, R3 = 1 kΩ

Measure the current across R1, R2, R3 and Voltage between node 2 and 7.

**Theoretical Calculations:**

- In a parallel circuit, the voltage across every branch equals the supply voltage.
- Voltage between node 2 and 7: V = 9 V
- Current through R1: I_R1 = V / R1 = 9 V / 10,000 Ω = 0.9 mA
- Current through R2: I_R2 = V / R2 = 9 V / 2,000 Ω = 4.5 mA
- Current through R3: I_R3 = V / R3 = 9 V / 1,000 Ω = 9.0 mA
- Total current (KCL): I_T = I_R1 + I_R2 + I_R3 = 0.9 + 4.5 + 9.0 = 14.4 mA ✓

**Theoretical Values:**

| R1 Current | R2 Current | R3 Current | Voltage (Node 2–7) |
|---|---|---|---|
| 0.9 mA | 4.5 mA | 9.0 mA | 9 V |

**Practical Values:**

| R1 Current | R2 Current | R3 Current | Voltage (Node 2–7) |
|---|---|---|---|
| 0.9 mA | 4.5 mA | 9.0 mA | 9 V |

*Note: In a parallel circuit, the voltage is identical across every branch. Branches with smaller resistance carry more current (Ohm's Law). Small differences from practical readings are due to resistor tolerances (±5%).*

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### Task 2: Parallel Circuit — Vs = 9V, R1 = 100 Ω, R2 = 3 kΩ, R3 = 5 kΩ, R4 = 10 kΩ

Measure the current across R1, R2, R3, R4 and Voltage between node a and e.

**Theoretical Calculations:**

- In a parallel circuit, the voltage across every branch equals the supply voltage.
- Voltage between node a and e: V = 9 V
- Current through R1: I_R1 = V / R1 = 9 V / 100 Ω = 90 mA

- Current through R2: $I_{R2} = V / R2 = 9\ V / 3{,}000\ \Omega = 3.0\ mA$
- Current through R3: $I_{R3} = V / R3 = 9\ V / 5{,}000\ \Omega = 1.8\ mA$
- Current through R4: $I_{R4} = V / R4 = 9\ V / 10{,}000\ \Omega = 0.9\ mA$
- Total current (KCL): $I_T = 90 + 3.0 + 1.8 + 0.9 = 95.7\ mA$ ✓

**Theoretical Values:**

| R1 Current | R2 Current | R3 Current | R4 Current | Voltage (a–e) |
|---|---|---|---|---|
| 90 mA | 3.0 mA | 1.8 mA | 0.9 mA | 9 V |

**Practical Values:**

| R1 Current | R2 Current | R3 Current | R4 Current | Voltage (a–e) |
|---|---|---|---|---|
| 90 mA | 3.0 mA | 1.8 mA | 0.9 mA | 9 V |

*Note: R1 = 100 Ω carries the largest current (90 mA) because it has the smallest resistance. The branch with the lowest resistance always draws the most current in a parallel circuit.*

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## Post Lab Questions

**1. Why is the value of voltage across each resistor in a parallel circuit the same?**

In a parallel circuit, all resistors share the same two nodes (connection points). Since voltage is defined as the potential difference between two points, any component connected between the same two nodes experiences the same potential difference. This is a fundamental consequence of Kirchhoff's Voltage Law (KVL): the voltage drop across each parallel branch must equal the supply voltage, because each branch forms an independent loop with the source. Therefore, $V_{R1} = V_{R2} = V_{R3} = V_s$ regardless of the resistance values.

**2. When parallel resistors are of three different values, which has the greatest power loss?**

The resistor with the SMALLEST resistance dissipates the greatest power. In a parallel circuit, the voltage across every branch is equal (V). Power dissipated by a resistor is given by $P = V^2 / R$. Since V is constant for all branches, power is inversely proportional to resistance — a smaller R gives a larger P. For example in Task 2: $P_{R1} = (9)^2 / 100 = 0.81\ W$, while $P_{R4} = (9)^2 / 10{,}000 = 0.0081\ W$. R1 (100 Ω) dissipates 100× more power than R4 (10 kΩ). The smallest resistor always has the greatest power loss in a parallel circuit.

**3. How does the current across any branch of a parallel circuit vary?**

In a parallel circuit, the current through each branch is inversely proportional to the resistance of that branch (Ohm's Law: $I = V/R$). Since the voltage is the same across all branches: a branch with a smaller resistance carries a larger current, and a branch with a larger resistance carries a smaller current. The total current from the source equals the sum of all branch currents (KCL): $I_{total} = I_{R1} + I_{R2} + I_{R3} + \ldots$ Adding more parallel branches always increases the total current drawn from the source.

**4. What happens to total resistance in a circuit with parallel resistors if one of them opens (i.e., removed)?**

If one parallel resistor opens (is removed or breaks), the total resistance of the circuit INCREASES. This is because the total parallel resistance is given by $1/R_{total} = 1/R1 + 1/R2 + 1/R3 + \ldots$ Removing a branch removes one term from the sum, making $1/R_{total}$ smaller, which means $R_{total}$ becomes larger. The remaining branches continue to operate normally — unlike a series circuit, the other resistors are unaffected. The total current from the source decreases because R_total increased. The voltage across the remaining resistors stays the same (equal to the supply voltage).

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*— End of Lab 4 Solved Manual — All theoretical values computed using Ohm's Law and Kirchhoff's Current Law (KCL).*